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Upsert rows in one statement

Oracle MERGE updates the rows that exist and inserts the ones that don't.

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Why it matters

A nightly price feed arrives. Some products already exist, some are new.

Without MERGE you write an UPDATE, then an INSERT for the rest. MERGE does both in one statement.

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The data

Target table: products

id	price
1	100
2	250

Source table: price_feed

id	price
2	275
3	90

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The MERGE

```
MERGE INTO products p
USING price_feed f
ON (p.id = f.id)
WHEN MATCHED THEN
  UPDATE SET p.price = f.price
WHEN NOT MATCHED THEN
  INSERT (id, price)
  VALUES (f.id, f.price);
```

ON true: update. ON not true: insert.

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The result

id	price	
1	100	untouched
2	275	updated
3	90	inserted

Product 1 is not in the feed, so MERGE leaves it alone.

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Add a WHERE to each branch

```
WHEN MATCHED THEN
  UPDATE SET p.price = f.price
  WHERE p.price <> f.price
WHEN NOT MATCHED THEN
  INSERT (id, price)
  VALUES (f.id, f.price)
  WHERE f.price > 0
```

The update WHERE can use source or target columns. The insert WHERE can use only source columns.

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Delete while you merge

```
WHEN MATCHED THEN
  UPDATE SET p.price = f.price
  DELETE WHERE (p.price = 0)
```

Only rows this MERGE updated can be deleted, and the check uses the new value.

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Two errors to know

ORA-30926: two source rows match the same target row. Make each source row match one target row, for example with GROUP BY.

ORA-38104: you tried to UPDATE a column used in the ON clause. Update other columns only.

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Recap

- ✔ MERGE INTO target USING source ON (...)
- ✔ WHEN MATCHED: UPDATE, WHEN NOT MATCHED: INSERT
- ✔ You need at least one of the two
- ✔ Optional WHERE on each branch
- ✔ 2 source rows, 1 target row: ORA-30926

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Save this for your next upsert

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